

Insider Threat Detection based on Knowledge Graph and Large Language Model

Yingna Li *

*The Third Research Institute of
Ministry of Public Security
The Third Research Institute of
Ministry of Public Security
Shanghai, China
Ziying219@126.com*

Zhiguo Ding

*The Third Research Institute of
Ministry of Public Security
The Third Research Institute of
Ministry of Public Security
Shanghai, China*

Zheng Yan

*Shanghai Public Security Bureau
Shanghai Public Security Bureau
Shanghai, China*

Zhuahua Li

*The Third Research Institute of
Ministry of Public Security
The Third Research Institute of
Ministry of Public Security
Shanghai, China*

Hang Shao

*The Third Research Institute of
Ministry of Public Security
The Third Research Institute of
Ministry of Public Security
Shanghai, China*

Abstract—This article proposes an insider threat detection method based on a combination of knowledge graph and large language model (LLM); first, the internal systems, users, IPs, access behaviors, etc. are modeled through the knowledge graph ontology; then, a few-shot learning information extraction method based on LLMs is used to extract knowledge from the behavior logs to complete the threat detection knowledge graph. Finally, the representation learning method based on the knowledge graph and the embedding based on the LLM are used to extract feature vectors, which are used as input of the insider threat detection model training based on Deep SVDD. The experimental results show that this method can automatically detect abnormal threat behaviors from massive logs at high accuracy, and has the ability to detect deeply hidden abnormal threat behaviors.

Keywords—insider threat detection, knowledge graph, large language model, deep-SVDD

I. INTRODUCTION

The internal network of enterprises is not only facing the risk of attack by external attackers, but also facing the threat from internal attackers. In general, an insider attacker generally refers to people within an organization, including current or former employees, contractors, and business partners [1]. Different from external attackers, internal attackers can use normal business processes to obtain information about internal networks, or use social relationships to obtain network administrator permissions, thus negatively affecting internal network security [2]. These potential threats are collectively referred to as "internal threats". The key event of internal threat from behind the scenes to the front stage is Snowden's "Prism gate" incident, which shows that internal threat has developed into an important issue in the field of network security [3]. There are three types of Insider Threats [4]:

- 1) Malicious Actions: These actions are typically carried out by disgruntled employees or individuals with harmful intentions. Examples include data theft, system sabotage, and misuse of privileged access.
- 2) Unintentional Actions: Unintentional insider threats arise from employees who lack security awareness or make mistakes. These may include accidental data sharing, mishandling of sensitive information, or failure to follow security protocols.
- 3) Covert Activities: Some insider threats are difficult to detect because they involve deliberate concealment. Attackers may slowly exfiltrate sensitive information or abuse legitimate privileges over time.

Abnormal behavior-based detection method is a most commonly used insider threat detection method, the method is based on big data and artificial intelligence technology. The method is data-driven, through big data analysis, feature extraction and classification of user behaviors are carried out to determine whether the user's behaviors are normal, and then possible abnormal behaviors are identified to determine whether they are internal threats [5][6]. From the perspective of algorithm model, the commonly used models for insider threat detection include Hidden Markov Models [7], Support Vector Machines, mixed Gaussian Model [8], Deep Learning methods [9] and Deep Recurrent Neural Networks [10], etc.

Current abnormal behavior detection method of internal users usually adopts the exception detection method that relies on rules set by experts, and the expert rules are usually summarized based on historical and past abnormal events. The main problems of this method are as follows: 1) it is difficult to conduct a complete and comprehensive summary of historical events manually, and a large part of abnormal events will be

missed; 2) The timeliness is not high, experts often need to find and summarize the same abnormal behavior after several times, so the lag is relatively serious; 3) Manual induction and summary can usually only realize the summary of a single event or a few highly correlated abnormal event patterns, and it is difficult to detect and summarize complex events associated with multiple users and multiple systems.

In view of the above shortcomings of existing methods, this paper proposes a deep anomaly detection method based on the combination of knowledge graph, representation learning and large language model. The powerful knowledge representation capability of knowledge graph is used to describe behavioral events and the correlation between them and users, systems, IP, etc., and the knowledge graph of security threat detection is automatically constructed by using large language model. Then, the representation learning technology based on attention mechanism is used to learn the vector features of the behavior event relationship network from the knowledge graph, and the semantic vector of the behavior event is generated based on the large language model. Finally, the deep anomaly detection model based on the neural network is trained.

II. SYSTEM ARCHITECTURE

The overall architecture of the thread detection system is shown in the in Fig. 1, which mainly includes five modules:

- 1) Threat knowledge graph ontology modeling: The knowledge graph is used to model user behaviors, systems, and network environments related to possible security threats and describe the relationships among them to provide basic support for subsequent abnormal behavior detection.
- 2) Knowledge graph construction based on large language model: Using few-shot learning based on LLM to extract entities, events and other knowledge from log data, and complete the transformation from original log data to knowledge in knowledge graph.
- 3) User behavior feature learning based on graph embedding: Based on the constructed threat knowledge graph, the graph embedding learning technology is used to convert the user behavior symbolized in the graph into the feature vector represented by distributed vectors.
- 4) Event vector feature transformation based on LLM: Using the semantic embedding ability of LLM, the knowledge in knowledge graph is converted into text description in the unit of events, and then converted into semantic vector by LLM.
- 5) Threat detection model training: The abnormal behavior detection model based on Deep SVDD is trained based on the feature vectors above to realize the discovery of security threats.

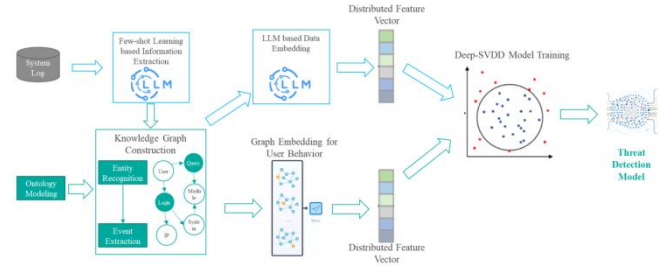


Figure 1. The architecture of the threat detection system.

III. SYSTEM IMPLEMENTATION

A. Ontology Modeling of Threat Knowledge Graph

Knowledge graph modeling is performed based on user behavior log data and security threat analysis goals. The entity types in the knowledge graph include “User”, “Department”, “System”, “System Module”, and “IP” etc.; the relationship types include “Department of User” and “System contains Modules” etc.; the event types include “Login”, “Query”, “Update”, and “Delete”. The graph mainly associates various types of entities through the "operational IP", "operational system and module" and other elements of the event, and finally forms a unified and associated knowledge graph. Since ontology modeling is very important to the detection model and the ontology scale is small, an approach based on artificial ontology modeling is adopted.

B. Construction of user behavior knowledge graph

After the ontology modeling is completed, the above types of entities, relationships and events need to be analyzed and extracted from the log data, and then filled into the knowledge graph to complete the knowledge graph. The common methods for entity and event extraction are rule-based method and deep learning-based method. The rule-based method has high accuracy but low recall, while the deep-learning-based method needs to label a large number of training corpuses. The rise of LLMs brings a new path for natural language processing. Based on the powerful language understanding and generation ability of LLM, knowledge extraction can be accomplished by few-shot learning based on LLM with a few sample prompts. Therefore, this paper adopts entity and event knowledge extraction method based on LLM. Taking event extraction as an example, the prompt structure adopted is as follows:

Task type: event extraction
 Event Type: System Access
 Event argument role: [access user, access time, access IP, expiration time]
 Log data input: [...]
 Please output the event extraction results:

C. User Behavior Feature Learning based on Graph Representation Learning

After the knowledge graph is constructed, the explicit features and deep correlation features of user behavior contained in the knowledge graph are studied. In this

paper, a method based on graph representation learning is adopted to realize feature learning in the user behavior graph, which can obtain a distributed representation based on dense vectors from knowledge representation learning based on discrete symbols, so as to better capture hidden knowledge. In this study, a neural network called Gated Attention Networks (GaAN) is used [11].

Different from the traditional multi-head attention mechanism (it consumes all attention heads equally), GaAN uses a convolutional sub-network to control the importance of each attention head. It assigns a different importance to each head by calculating an additional soft gate between 0 (low importance) and 1 (high importance). Combining multi-head attention aggregators, we get the formula for gated attention aggregators:

$$\mathbf{y}_i = \text{FC}_{\theta_o}(\mathbf{x}_i \oplus \bigoplus_{k=1}^K (g_i^{(k)} \sum_{j \in \mathcal{N}_i} w_{i,j}^{(k)} \text{FC}_{\theta_v^{(k)}}(\mathbf{z}_j))),$$

$$\mathbf{g}_i = [g_i^{(1)}, \dots, g_i^{(K)}] = \psi_g(\mathbf{x}_i, \mathbf{z}_{\mathcal{N}_i}), \quad (1)$$

Where $g_i^{(k)}$ is a scalar gate value at the Kth header of node I; in order to ensure that the added gate does not introduce too many additional parameters, a convolutional network ψ_g is used here, which takes the characteristics of the center node and adjacent nodes to generate gate values. ψ_g has multiple approaches involved.

D. Event Feature Vector Generation based on LLM

The events of threat knowledge graph are mainly described in text and symbolic way, the feature information contained in language and symbols can also be captured by semantic vectorization capability based on LLM. As a result of learning language features from data on the Internet-scale, the LLM's ability to understand language has basically approached that of human beings. By converting the user's text input into high-dimensional distributed vectors, the transformation from text to distributed feature vectors can be realized. The implementation process is very simple, the semantic vector model receives the user input in the form of text and then outputs the distributed vector. After testing and evaluation, the final semantic vector-LLM adopted in this paper is the version of sgpt-bloom-1b7 [12][13].

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E. Threat Detection Model Training based on Deep Neural Network

After the feature vectors are obtained, the anomaly detection model is trained. Since the pattern of users' abnormal behavior is constantly changing, it is difficult to obtain the corpus needed by the anomaly detection model to train the model through manual annotation, so an unsupervised method is needed. Here the anomaly detection model training using Deep SVDD model 错误!未找到引用源。 . It is a single classification anomaly detection method, which does not require the label of each training sample, but only needs to ensure that the vast majority of samples are normal samples, and describes the behavior of normal data by learning a "hypersphere".

The principle of Deep SVDD is shown in Fig. 2. By trying to learn a neural network, the data points are mapped to another space, where there is a hypersphere with c as the center and R as the radius. The neural network needs to ensure that most points (normal points) in the new space fall within the sphere, while a small number of points (abnormal points) fall outside the sphere. Then the anomaly is detected.

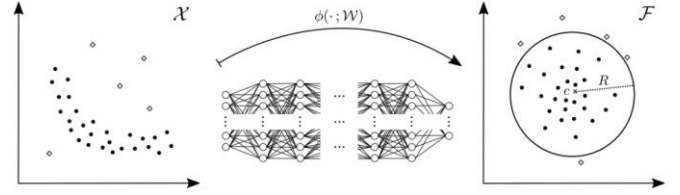


Figure 2. The principle of Deep SVDD

Finally, in the case of anomaly detection, in order to illustrate the relationship between the behavior chain and its related logs by using graph model, a variant model called GRU-based Deep SVDD is adopted, which sorts all events in a call chain according to the occurrence time, and trains the Deep SVDD model based on this sequence.

IV. EXPERIMENT

In order to verify the threat detection method proposed in this paper, the system access log data of an enterprise within a period of time is used as the test data set. After filtering the incomplete data, 998,273 logs were selected, including 1,082 employees, 17 departments, 1,872 IP addresses, 26 systems, and 388 system modules, involving 27 types of system operations. This log data contains 382 abnormal behavior log data that has been manually confirmed, of which 335 are the abnormal behavior of a single user in a single system(Simple), and the other 47 involve multiple users or multiple systems(Complex). Examples of log data are show in Table 1:

TABLE I. EXAMPLES OF LOG DATA.

Time	IP	Action	System	Module	User
2021-12-01 12:36:49	172.20. 9.22	login	OA	Login	alphas
2021-12-01 12:38:08	172.20. 9.22	Approve leave request	OA	Attendan ce	alphas
2021-12-01 12:38:08	172.20. 9.22	Submit purchase order	ERP	Purchasi ng	alphas

2021-12-01 12:40:22	172.20. 9.22	Submit purchase order	ERP	Purchasing	alphas
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The verification results of the threat detection model trained on the selected test data set are shown in Table 2.

TABLE 2. EXPERIMENT RESULT.

	Total	Simple abnormal behavior	Complex abnormal behavior
detected	422	351	71
correct	354	309	45
Precision	0.839	0.88	0.634
Recall	0.923	0.922	0.957

The results show that the proposed model has good abnormal behavior detection ability, the overall accuracy is 83.9%, and the recall is 92.3%. In particular, the model has the ability to detect complex abnormal behavior, although the accuracy is only 63.4%, but the recall reached 95.7%. According to the analysis, the reason why the recall of the model is relatively high is that in the threat detection scenario, it is more important to fully identify potential threats than to accurately identify them, because the abnormal behaviors detected are usually manually confirmed and corresponding interventions are taken, even if the identification is wrong, it will not cause harm; but missing undetected behavior often creates more of a threat.

V. CONCLUSION

In this paper, a method based on knowledge graph modeling is proposed to represent user log behavior, which can effectively describe individual operation behavior information and also describe the correlation between behaviors, so as to realize the joint abnormal event detection between multi-systems and multi-users. At the same time, this paper adopts knowledge extraction method based on LLM, which greatly reduces the cost of knowledge graph construction. In the stage of security threat detection, by using the graph representation learning method based on the gated attention mechanism and the semantic vector generation based on the LLM to achieve the capture of anomaly detection behavior features, we can effectively obtain the individual behavior features, association behavior features and hidden features in the graph structure.

Finally, a Deep anomaly detection model based on neural network, GRU-based Deep SVDD, is adopted. This model is an unsupervised training model, which is suitable for the scene in this project where the abnormal behavior pattern cannot be marked manually. On the other hand, the model can receive distributed vector features of graph representation learning and large model semantic vectorization as inputs to capture deep hidden features. Meanwhile, event time series features are used in the training process to further improve the performance of the model.

Future works will focus on using LLMs with strong logical reasoning ability such as OpenAI o1 [15] to conduct in-depth analysis and reasoning based on the patterns of user behavior, so as to find abnormal situations.

REFERENCES

- [1] Ivan Homoliak, Flavio Toffalini, Juan Guarnizo, Yuval Elovici, Martín Ochoa: "Insight into Insiders and IT: A Survey of Insider Threat Taxonomies, Analysis, Modeling, and Countermeasures", 2018, ACM Computing Surveys (CSUR). Vol. 52 (2019), Issue 2; arXiv:1805.01612. DOI: 10.1145/3303771.
- [2] GUO Shize, ZHANG Lei, PAN Yu, et al. Survey on Insider Threat Detection Method[J]. Journal of Data Acquisition and Processing, 2022, 37(03):488-501.
- [3] CLAYCOMB W R, NICOLL A. Insider threats to cloud computing: Directions for new research challenges[C]// Proceedings of the IEEE 36th Annual Computer Software and Applications Conference (COMPSAC). Lzmir, Turkey: IEEE, 2012: 387-394.
- [4] Chattopadhyay, A., & Capretz, L. F. (2021). Insider Threat Detection in Cyber Security: A Systematic Review. Computers & Security, 109, 102355.
- [5] AGRAFIOTIS I, EROLA A, GOLDSMITH M, et al. A tripwire grammar for insider threat detection [C] // Proceedings of the 8th ACM CCS International Workshop on Managing Insider Security Threats. New York: ACM Press, 2016: 105-108
- [6] MORIANO P, PENDLETON J, RICH S, et al. Insider threat event detection in user-system interactions [C]// Proceedings of the 2017 International Workshop on Managing Insider Security Threats. Dallas, Texas, USA: ACM, 2017: 1-12.
- [7] RASHID T, AGRAFIOTIS I, NURSE J R. A new take on detecting insider threats: Exploring the use of hidden Markov models[C]// Proceedings of the 8th ACM CCS International Workshop on Managing Insider Security Threats. New York: ACM Press, 2016: 47-56.
- [8] TABASH K A, HAPPA J. Insider-threat detection using gaussian mixture models and sensitivity profiles[J]. Computers & Security, 2018, 77: 838-859.
- [9] SOH C, YU S, NARAYANAN A, et al. Employee profiling via aspect-based sentiment and network for insider threats detection[J]. Expert Systems with Applications, 2019, 135: 351-361
- [10] ZHANG D, ZHENG Y, WEN Y, et al. Role-based log analysis applying deep learning for insider threat detection[C]// Proceedings of the 1st Workshop on Security-Oriented Designs of Computer Architectures and Processors. New York: ACM Press, 2018: 18-20.
- [11] Jiani Zhang, Xingjian Shi, Junyuan Xie, Hao Ma, Irwin King, Dit-Yan Yeung: "GaAN: Gated Attention Networks for Learning on Large and Spatiotemporal Graphs", 2018; arXiv:1803.07294.
- [12] Niklas Muennighoff: "SGPT: GPT Sentence Embeddings for Semantic Search", 2022; arXiv:2202.08904.
- [13] <https://huggingface.co/bigscience-data/sgpt-bloom-1b7-nli>
- [14] Lukas Ruff, Robert Vandermeulen, Nico Goernitz, Lucas Deecke, Shoaib Ahmed Siddiqui, Alexander Binder, Emmanuel Müller, Marius Kloft. Deep One-Class Classification[C], Proceedings of the 35th International Conference on Machine Learning, PMLR 80:4393-4402, 2018. <https://ls9-www.cs.tu-dortmund.de/publications/ICML2018.pdf>
- [15] <https://openai.com/index/introducing-openai-o1-preview/>